Limits to Arbitrage in the Indian Equity Market

Stage I: Theory, Data and the Futures-Cash Band

BS Project, Stage I (EC 495)

Prince Kumar

Roll No. 23B3032, BS Economics

Supervisor: Prof. Aditi Chaubal

Department of Economics
Indian Institute of Technology Bombay

November 2026



Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Motivation

- A share and a future on it end in the same place, so the futures price should equal the share price plus the cost of carrying it to expiry. The gap is the basis.
- Rich future: buy the share, sell the future (**cash-and-carry**); cheap future: the reverse, which needs a short sale.
- In practice arbitrage is limited: it costs money, needs capital, carries risk and often needs a short position (Shleifer and Vishny, 1997; Gromb and Vayanos, 2010).
- India changes almost every such limit on a known date:
  - the securities transaction tax (STT) on futures rose in 2023, 2024 and 2026;
  - only F&O stocks can be shorted;
  - position limits are set by formula and were redesigned in October 2025.
- India also has an unusual rule, the F&O ban, which no study has examined.

The F&O ban

- Market-wide open interest above **95%** of the market-wide position limit (MWPL): ban from the next day
- **Everyone**, not only arbitrageurs, may only reduce positions
- Ban ends when open interest falls to **80%**
- 1,788 spells in 141 stocks, 2015 to 2026

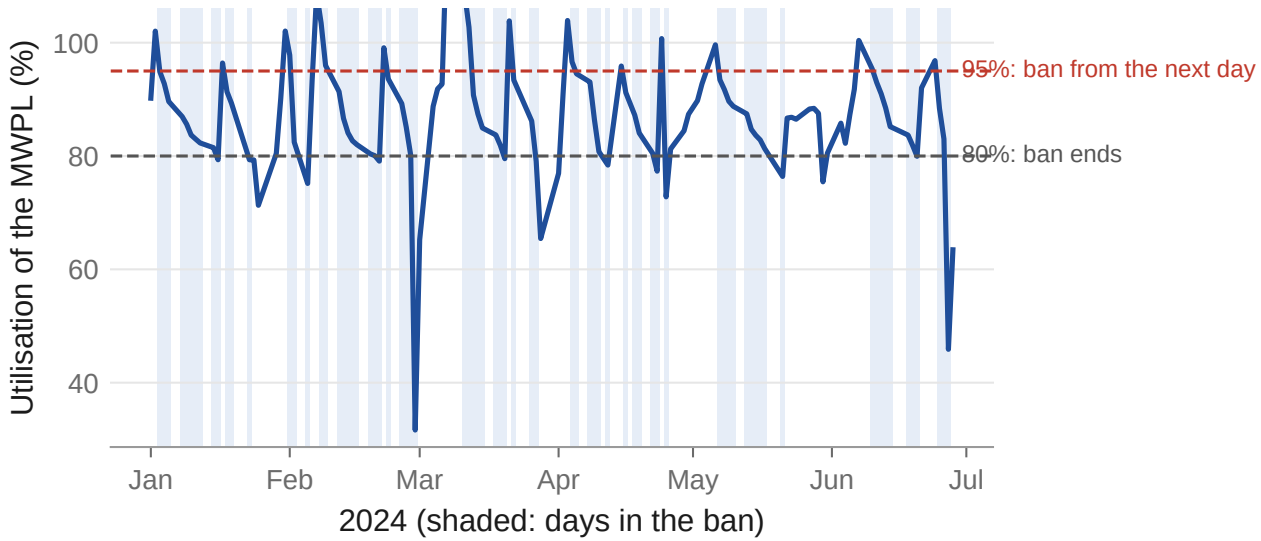


Figure 1: Utilisation of the MWPL for SAIL, January to June 2024; shaded: ban days.

Research question and the two stages

Research question

How strongly do limits to arbitrage bind in the Indian equity market, and how much mispricing do they explain?

Stage I (this presentation)

1. A model with every Indian limit as a parameter
2. A test plan frozen before any outcome was related to any treatment
3. A data set from public exchange files
4. How wide is the no-arbitrage band?
5. Do costs and frictions set its edges?
6. Does the ban bind on trading?

Stage II

1. What a ban does to the basis
2. Position limits before a ban
3. Arbitrage capital and the band
4. Short-sale access across the market
5. Policy

Research gaps

Table 1: Earlier Indian estimates of the futures-cash band

Study	Sample	Data	Band
Aggarwal (2015)	141 stocks, 2013	One-second	-15 to $+3$ bp
Shankar, Sankar and Kiran Kumar (2019)	102 stocks, 2007 to 2009	Five-minute	-39 to -10 bp
Slivka, Shah and Wu (2011)	Cost calculation	Fee schedules	Twice mature markets
This study	**360 stocks, 2015 to 2026**	**Daily**	$-$**34.5** to $+$**23.9** bp

- Earlier bands are intraday objects over a few months or years; the overnight band, where delivery taxes are paid, has not been measured over a long panel.
- No study has modelled or tested the F&O ban.

Contributions of Stage I

1. A model of arbitrage in which each Indian limit is a parameter, including the ban as a reduce-only rule for every participant.
2. A pre-analysis plan with falsifiers, frozen with a time stamp before the tests.
3. The no-arbitrage band of Indian stock futures over twelve years of daily data, with threshold and smooth-transition models.
4. A test of whether costs and frictions set the band.
5. The first evidence on how the F&O ban affects trading, and how the 2025 rules changed the number of bans.

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Why arbitrage is limited

- **Capital:** arbitrageurs invest other people's money; losses bring withdrawals when gaps widen (Shleifer and Vishny, 1997).
- **Risk:** noise-trader risk deters arbitrageurs with short horizons (De Long et al., 1990).
- **Costs and short sales:** a shorting cost makes the band around fair value asymmetric; margins make capital bind (Gromb and Vayanos, 2010).
- **The band:** threshold models of futures and cash (Dwyer, Locke and Yu, 1996); smooth adjustment when arbitrageurs differ (Tse, 2001); costs estimated from the band (Ters and Urban, 2020).
- **India:** intraday bands for stock futures (Aggarwal, 2015; Shankar, Sankar and Kiran Kumar, 2019).

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Dated changes to the limits, 2015 to 2026

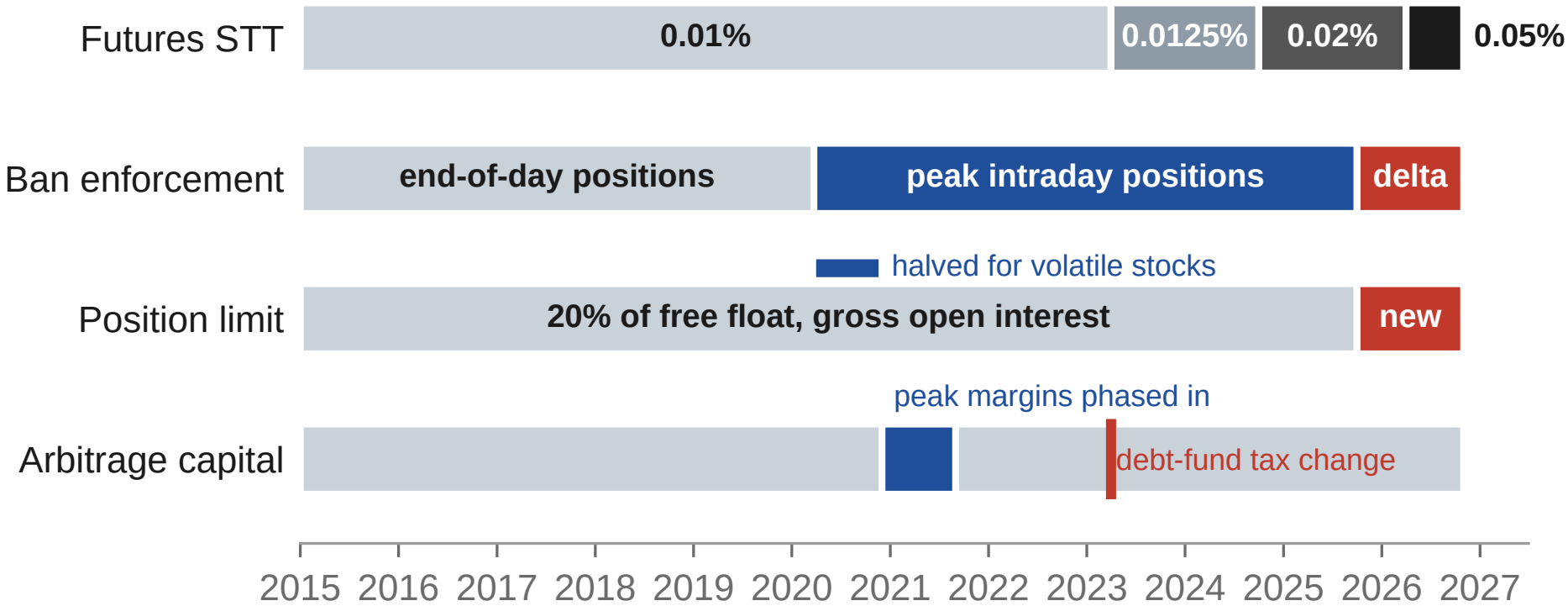


Figure 2: Futures STT, enforcement of the ban, the position-limit rule and arbitrage capital.

- Ban violations were judged on end-of-day positions to 22 March 2020, on peak intraday positions from 23 March 2020, and on end-of-day delta from October 2025.

Position limits, short selling and costs

Table 2: The market-wide position limit over the sample

Period	MWPL	Open interest tested
To 30 Sep 2025	20% of free float	Gross, futures and options
Mar to Nov 2020	Halved for volatile or heavily used stocks	Gross
From 1 Oct 2025	Lower of 15% of free float and 65 times average daily delivery; floor 10%	Delta-adjusted

- Only F&O stocks may be sold short; naked shorting is barred; stock lending is thin. The practical overnight short is a future, so reverse arbitrage is costly.
- Futures STT: 0.01% (2013), 0.0125% (Apr 2023), 0.02% (Oct 2024), 0.05% (Apr 2026); cash delivery pays 0.1% on each side.

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Set-up

- Basis $m = \ln F - \ln F^*$, with $F^* = (S - \mathrm{PV}(D))\, e^{r(T-t)}$.
- Outside traders create demand pressure u toward futures; an arbitrage sector holds a units of cash-and-carry (long stock, short future). Market clearing:

$$m = \frac{u - a}{\Phi}.$$

- The sector maximises

$$a\,m - \tfrac{\gamma}{2} a^2 - c_+ a^+ - c_- a^-, \qquad \gamma = \alpha \sigma^2,$$

  where c_+ is the cost of cash-and-carry and c_- the cost of the reverse trade.
- Every Indian limit is a parameter: costs ($c_\pm$), short sales (c_-), capital and risk (γ), position limits (the reduce-only rule).

The no-arbitrage band

Proposition 1 (An asymmetric band)

No arbitrage pays while m lies in $[-c_-, c_+]$; inside, m follows demand ($\partial m/\partial u = 1/\Phi$); outside, arbitrage absorbs part of each push ($\gamma/(\gamma\Phi + 1)$).

Proposition 2 (What moves the band)

Costs set the edges one for one; capital and risk set how far m travels beyond them; the band narrows toward expiry and sits below zero when $c_- > c_+$.

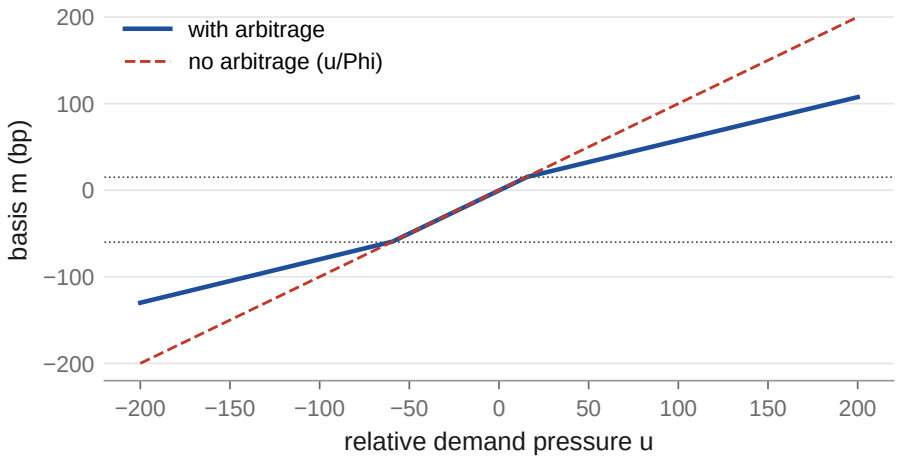


Figure 3: Equilibrium basis against demand pressure (illustrative costs).

Existing positions move the band

Proposition 3 (The band moves with the existing book)

With a position a_{t-1} and an unwinding cost c_u, the sector trades only when m_t leaves $[\gamma a_{t-1} - c_u,\ \gamma a_{t-1} + c_+]$.

- The edges shift with the size of the book, so a fixed-threshold model estimates an **average** edge, not the cost.
- Checked on data simulated from the model: with costs of 15 bp to add and 10 bp to unwind, the estimated thresholds were about -10 and 0 bp.
- Hence the cost test compares changes in the upper edge around tax increases, not its level.

The F&O ban as a reduce-only rule

Assumption 1 (Ban)

(B1) Arbitrageurs may only reduce: $0 \leq a_t \leq a_{t-1}$. *(B2) New futures demand is blocked; a share* χ *moves to the stock, the rest is withdrawn.*

Proposition 4 (One-sided correction)

Rich futures are no longer corrected; cheap futures are, while positions last.

Proposition 5 (Level effect)

The ban raises the basis when pressure comes from closing trades and lowers it when new demand keeps arriving.

Proposition 6 (Trading in a ban)

If the rule binds on every trade, each trade closes positions: $\Delta OI = -V$, *and futures volume falls.*

Stage I tests the trading proposition; the price propositions are tested in Stage II.

A simulated ban

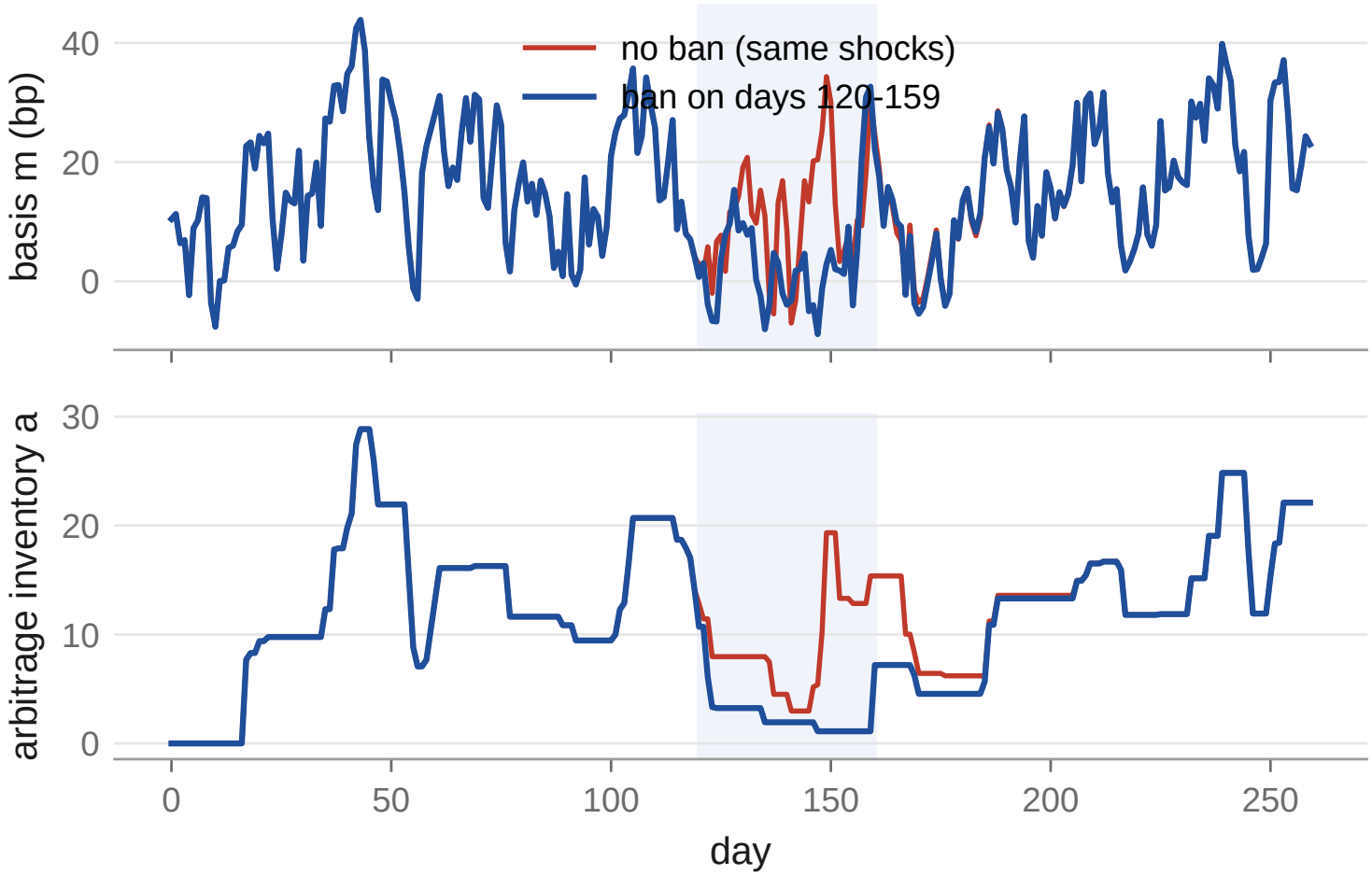


Figure 4: Same shocks with and without a ban on days 120 to 159: basis (top) and arbitrage position (bottom). Illustrative parameters.

Predictions and falsifiers

Table 3: Predictions of the model; primary hypotheses of the test plan in bold

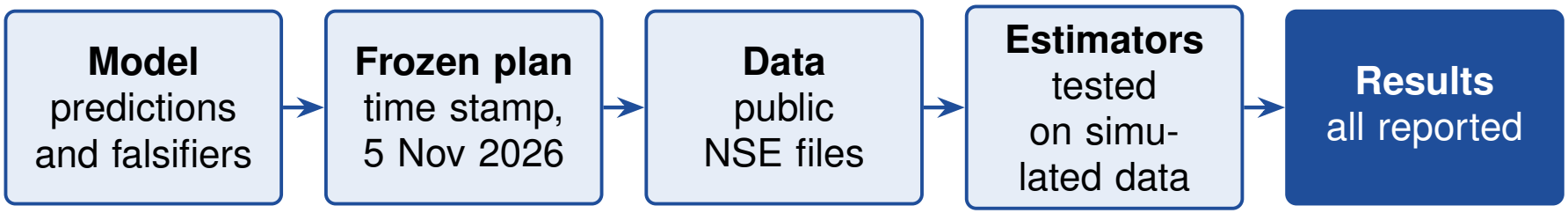
Label	Prediction	Contradicted if	Stage
Band	The basis moves in an asymmetric band	No band; symmetric	I
H3	Tax increases raise the upper edge	Edges unrelated to costs	I
H1	Wider bands where frictions are higher	No relation	I
H8	In a ban futures volume falls and $\Delta OI \approx -V$	Futures activity unchanged	I
H7	In a ban rich futures revert more slowly	No asymmetry	II
Level	The level effect follows the composition of trading	Sign unrelated	II
H6	Short-sale access speeds price adjustment	No change	II
H9, H4, H5, H2	Anticipation; capital; anomalies; bad news in bans	No relation	II

The four primary hypotheses are corrected jointly (Holm, 5%); secondary tests are unadjusted.

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Research design



- **Methods in Stage I:** three-regime threshold and smooth-transition models; event comparisons with placebo dates; two-way fixed effects with clustered errors; density test and regression discontinuity at 95%.
- Every detail the plan left open was fixed in a dated log before the test that needed it.

Data

Table 4: Public data sources

Source	Content	Coverage
NSE open interest, ban lists	MWPL, open interest, ban status	2010 to 2026
NSE bhavcopies, delivery files	Futures and cash prices, volumes, delivery	2015 to 2026
NSE corporate actions	Dividends and other events	2014 to 2026
NSE securities lending	Daily lending trades	2015 to 2026
OECD via FRED	Three-month interbank rate	2011 to 2026

Checks

Next-day flags match the official ban list for 11,753 of 11,764 flags; the 95% and 80% rules hold for 99.4% and 99.7% of spells; a full rerun reproduces every number.

The futures-cash basis

$$m_{it} = \ln \bar{F}_{it} - \ln\left(\bar{S}_{it} - \mathrm{PV}_{it}(D)\right) - r_t \tau_{it}$$

- Volume-weighted prices, not settlement prices (set at cost of carry on thin days)
- Near-month contract, at least 5 contracts, no dividend in its life
- 486,040 stock-days, 371 stocks
- Median $+1.1$ bp; 10th and 90th percentiles -39.1 and $+26.8$ bp

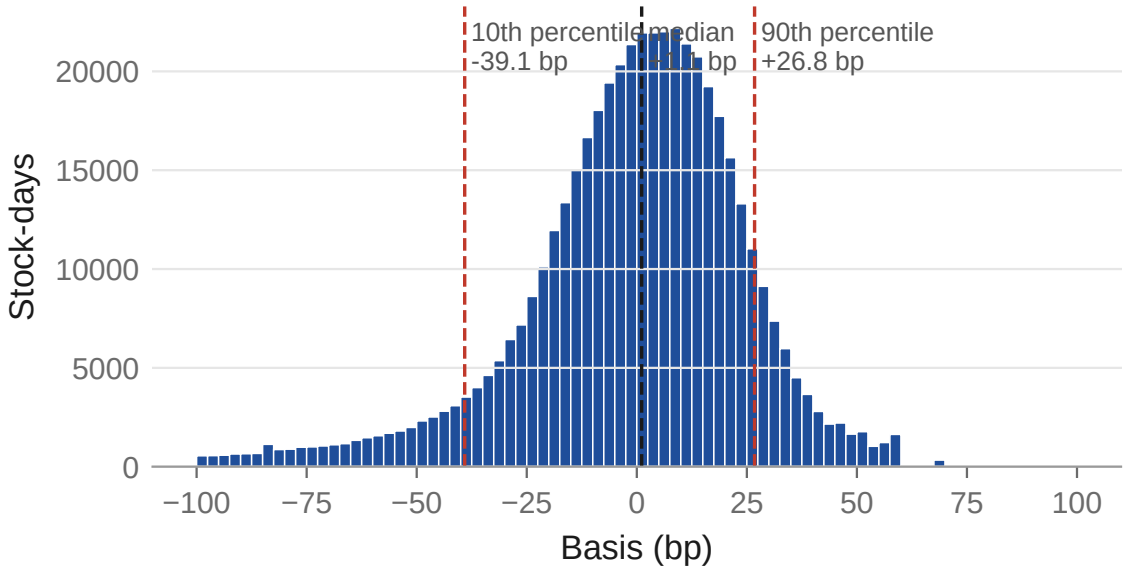


Figure 5: Distribution of the basis (bp).

Bans over time

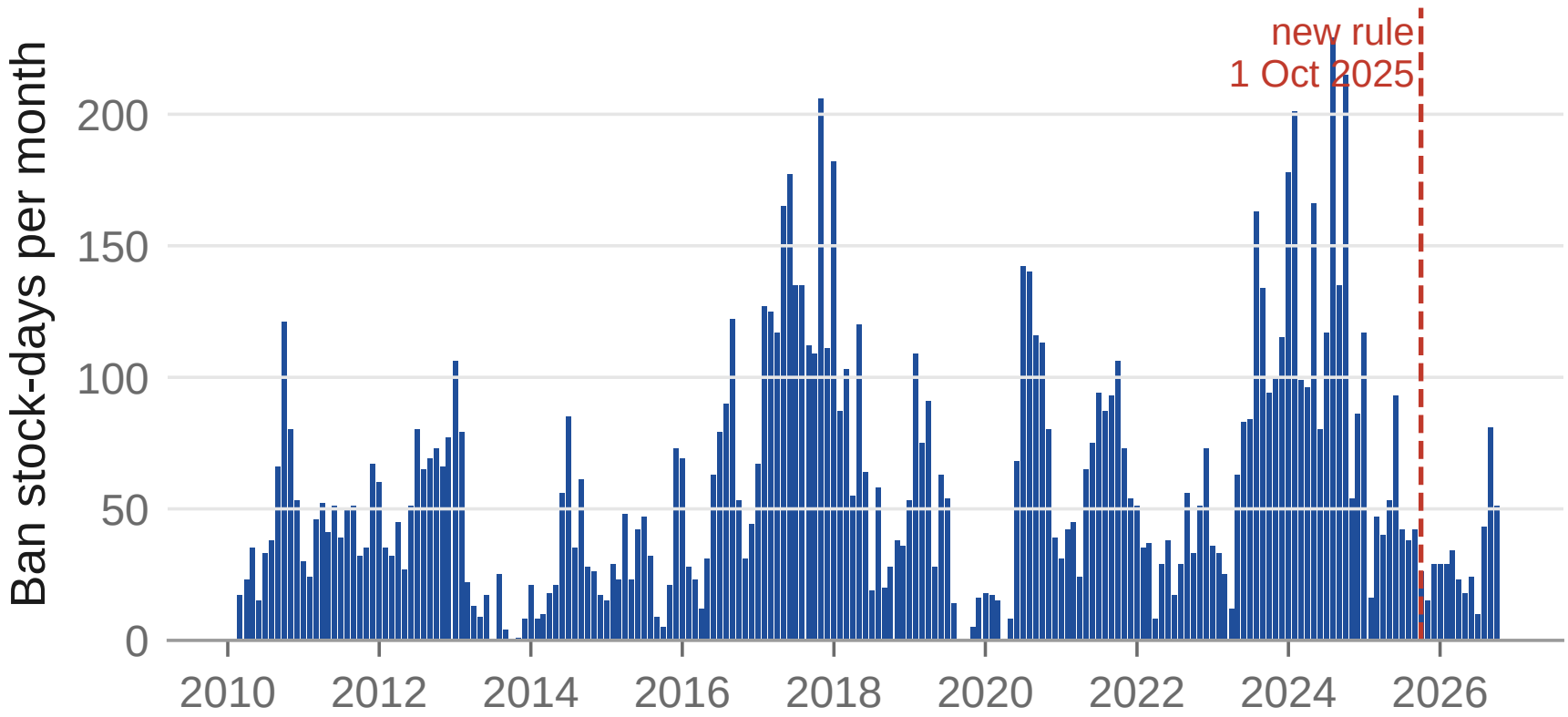


Figure 6: Stock-days in the ban per month, 2010 to 2026.

- 1,788 spells, 9,416 stock-days, 141 stocks (2015 to October 2026); a typical spell lasted 4 trading days before the reform and 9 after it.

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

A three-regime threshold model

$$\Delta m_t = a_r + \rho_r \, x_{t-1} + e_t, \qquad r = \begin{cases} \text{lower} & x_{t-1} < g_L \\ \text{middle} & g_L \le x_{t-1} \le g_U \\ \text{upper} & x_{t-1} > g_U \end{cases}$$

- x_{t-1}: yesterday's basis minus the stock's median on non-ban days; Δm_t: change of the same contract.
- Thresholds by least squares over a grid of percentiles (at least 10% in each outer regime).
- Linearity tested with a sup-F statistic and a wild bootstrap clustered by day; threshold intervals by resampling whole stocks.
- Prediction: ρ near zero inside the band, clearly negative outside.

The basis moves in a wide, asymmetric band

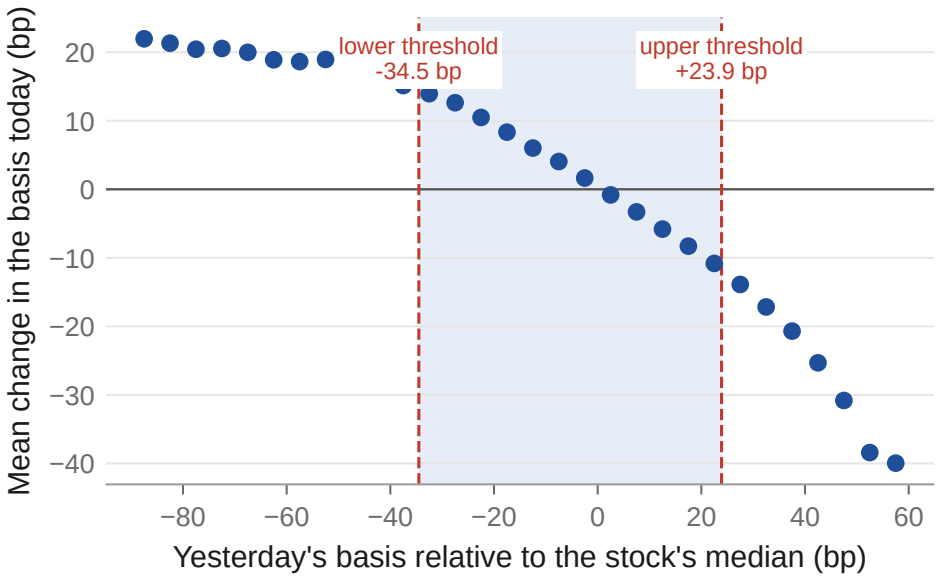


Figure 7: Mean daily change by yesterday's deviation.

Table 5: Pooled threshold model, 473,110 obs.

	Lower	Middle	Upper
Threshold	−34.5		+23.9
Share	0.12	0.76	0.12
Slope	−0.084	−0.468	−0.809
IV slope	−0.121	−0.192	−0.141

Linearity rejected (sup-F 7,471, critical value 31). The lower edge is further away, as expected when the reverse trade is costlier.

Noise, smooth adjustment and stability

- **Noise:** measurement error biases slopes toward -1; instrumenting x_{t-1} with x_{t-2} shrinks all three slopes. Where the band lies is robust; how fast the basis returns is not.
- **Smooth transition:** a logistic model fits almost as well as the threshold model; a symmetric exponential one fits worse.
- **Expiry:** lower edge -40 bp far from expiry, -28 bp in the last ten days.

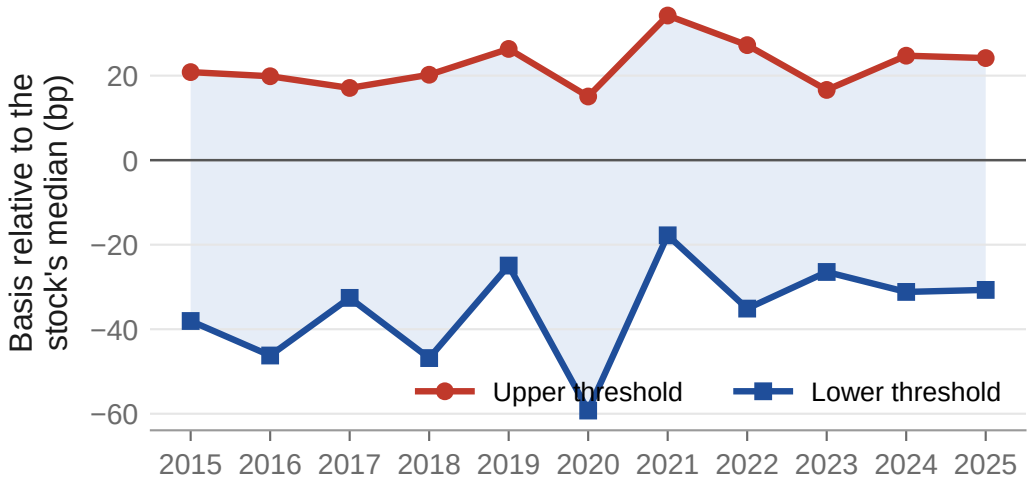


Figure 8: Thresholds by year, 2015 to 2025.

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Tax increases and the upper edge (H3)

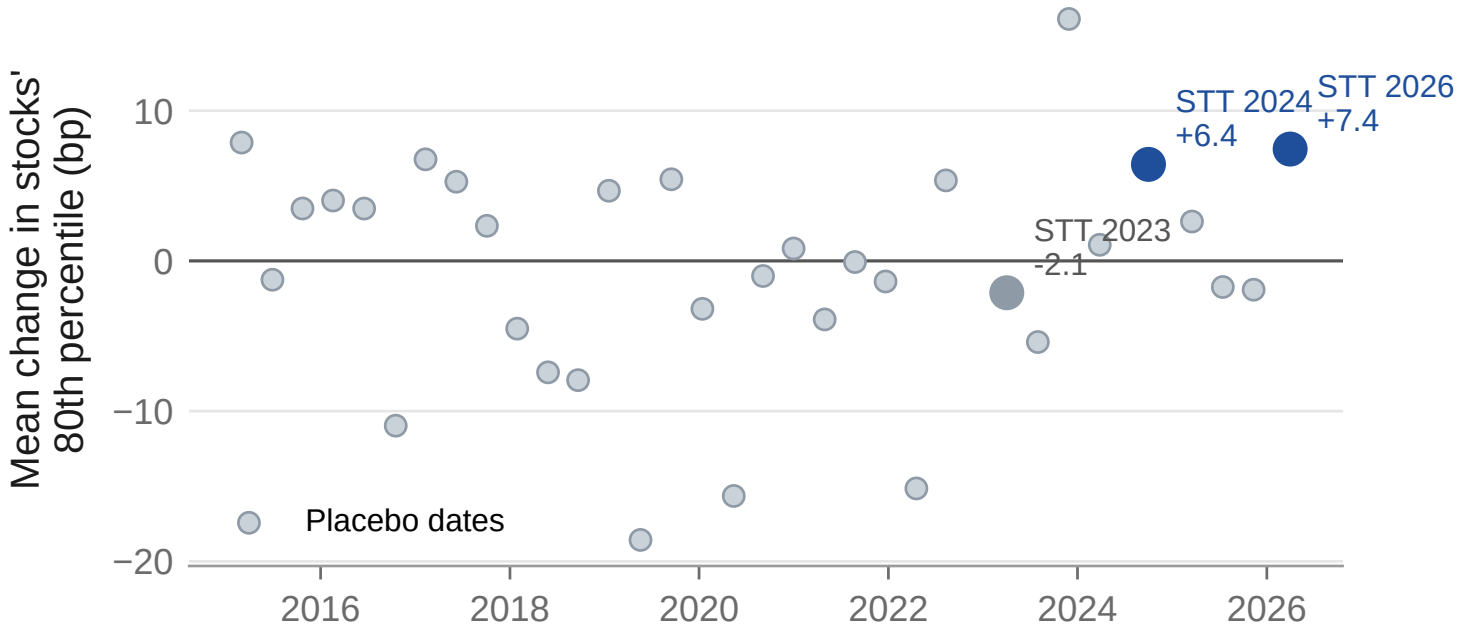


Figure 9: Mean change in stocks' 80th percentile of the basis at the STT dates and at 30 placebo dates (bp).

- Pre-specified: +6.96 bp (s.e. 0.61) pooling 2024 and 2026; ordered as the cost changes. Supported, and significant after Holm.
- But the rises exceed the cost changes, the median moved too, and placebo dates move as much (s.d. 7.4 bp).

The cost of a new cash-and-carry position

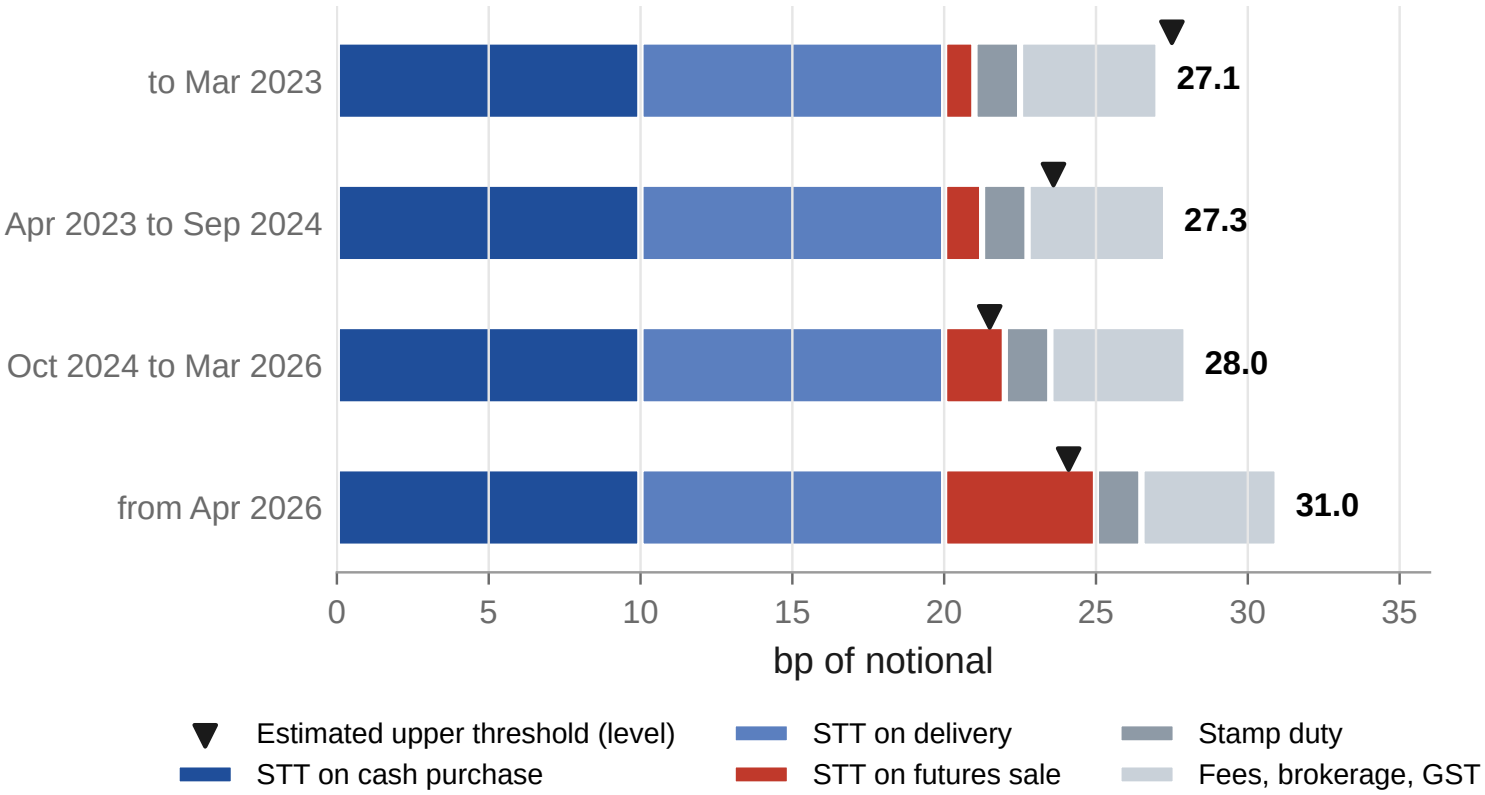


Figure 10: Round trip of an overnight position held to delivery, by STT period (bp of notional).

- A new round trip costs 27 to 31 bp, above the estimated upper edge (+22 to +28 bp); rolling costs 4 to 8 bp.
- New capital enters only at large gaps; funds that roll lean against small ones.

Frictions across stocks (H1)

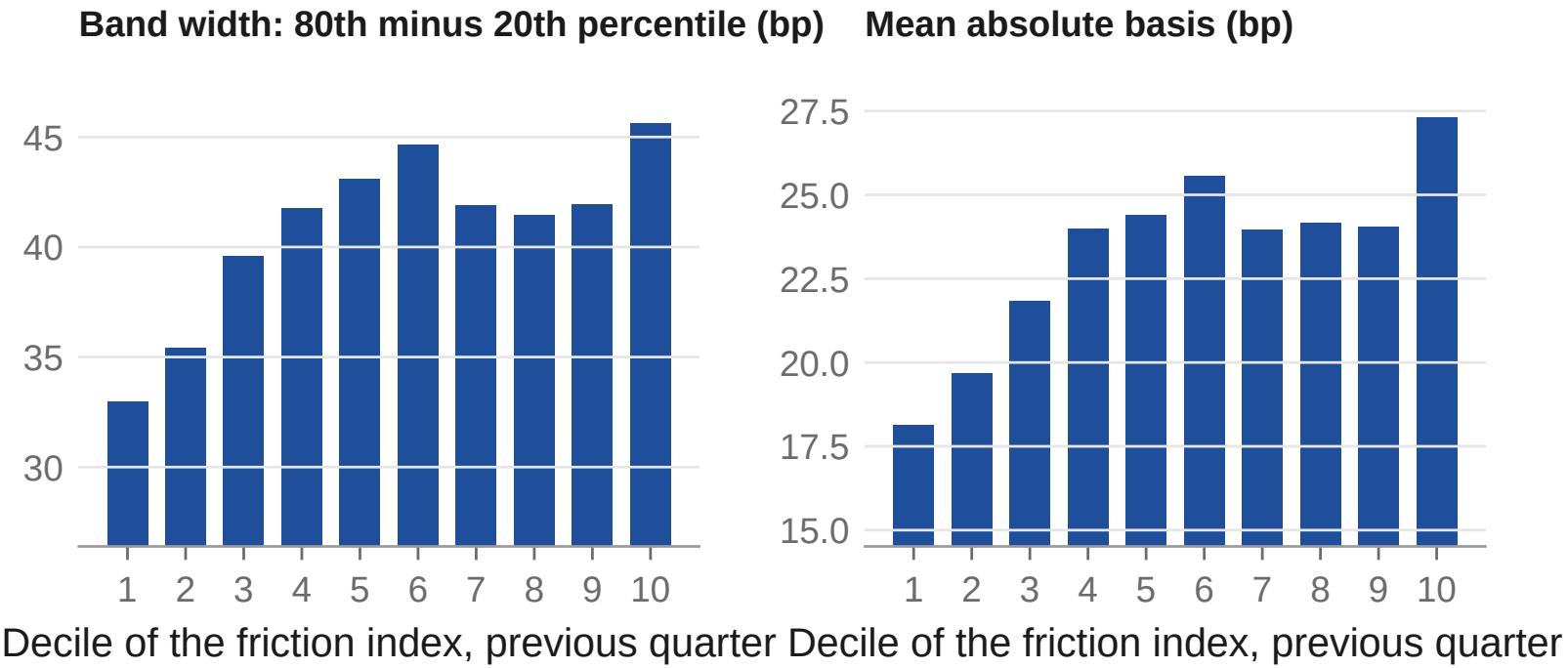


Figure 11: Band width and mean absolute basis by decile of a friction index of the previous quarter.

- Index: spread, illiquidity, idiosyncratic volatility, utilisation, lack of stock lending. One s.d. more: band $+3.5$ bp (s.e. 1.1, $p = 0.002$), mean width 40.8 bp.
- Across stocks only: with stock fixed effects the relation disappears.

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Trading in a ban (H8)

$$y_{it} = \beta \, \mathrm{Ban}_{it} + \alpha_{i,b(\omega_{i,t-1})} + \lambda_t + \varepsilon_{it}$$

Table 6: Ban days against the same stock's non-ban days at 80 to 95% utilisation, 2015 to September 2025

Outcome	Effect (s.e.)	Non-ban mean
log(1 + futures contracts)	−2.00 (0.11)	8.50
log(1 + cash shares)	+0.073 (0.027)	15.80
Delivery share	+0.041 (0.005)	0.26

- Futures trading falls by about 87%; cash trading rises and moves toward delivery. Supported, and significant after Holm.

Enforcement decides whether the rule binds

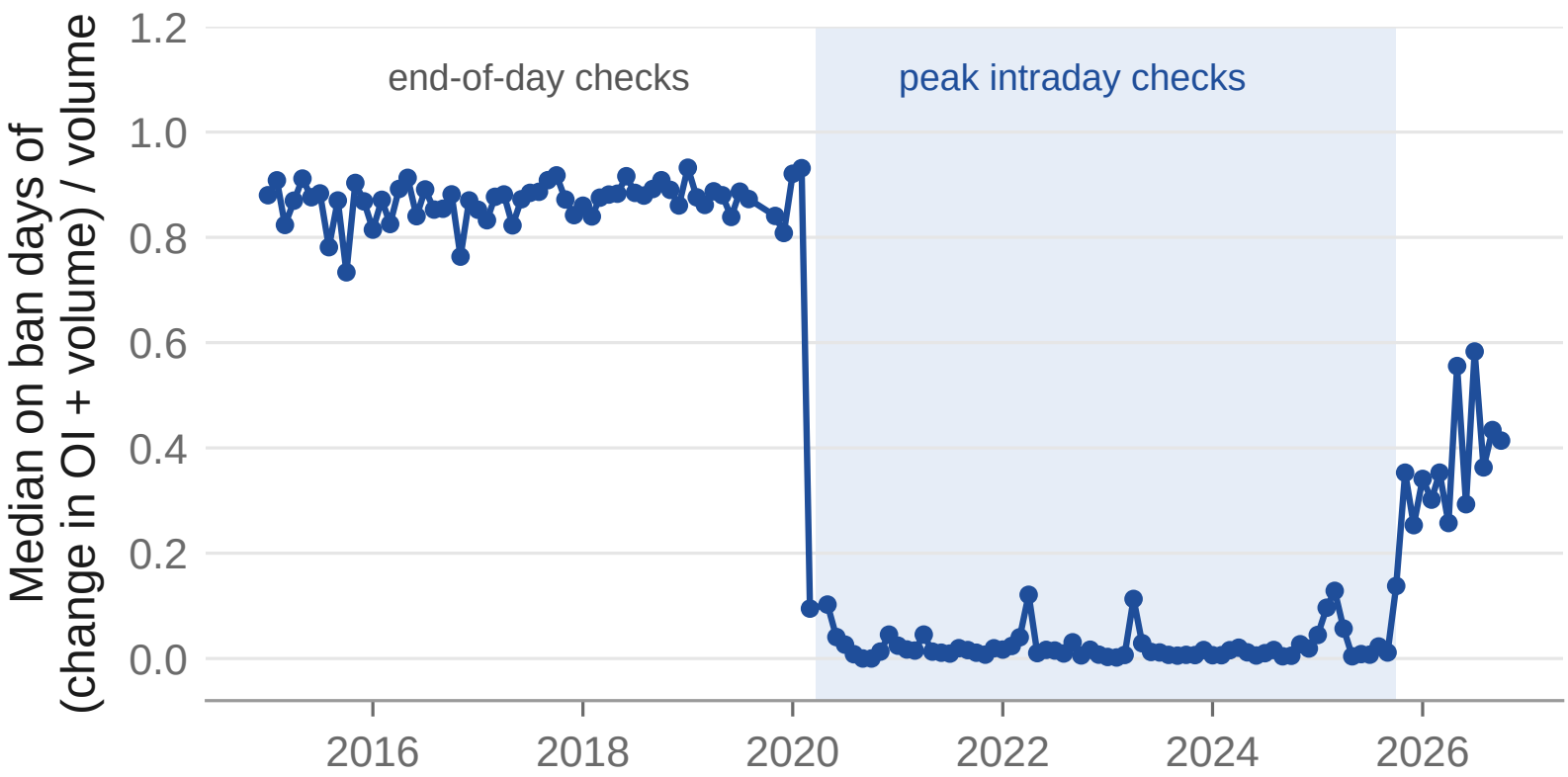


Figure 12: Monthly median of $R = (\Delta OI + V)/V$ on ban days; $R = 0$ when every trade closes a long and a short.

- End-of-day checks (to 22 March 2020): median 0.87. Peak intraday checks: 0.011. Delta rule (from October 2025): 0.19 to 0.36. Futures effect -0.97 vs -2.72 log points.

The 95% trigger

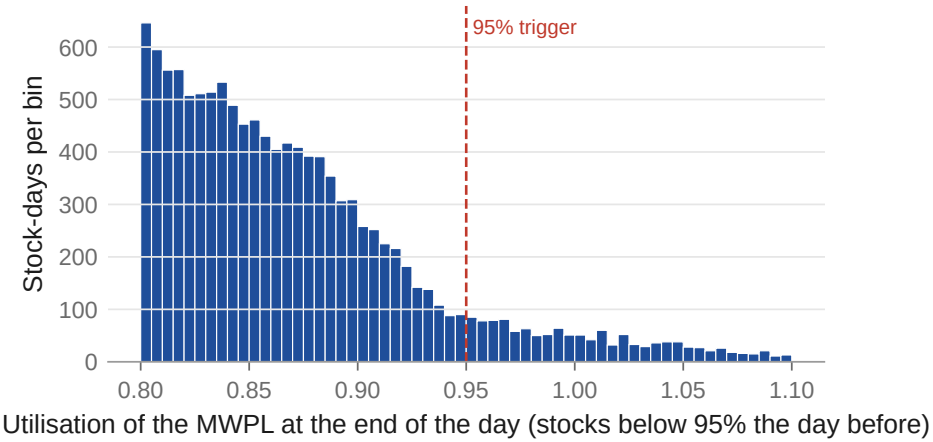


Figure 13: Utilisation of stocks below 95% the day before.

No bunching: $T = -0.18$, $p = 0.86$. The chance of crossing next day rises from 7% at 85% to about half at 93 to 95%.

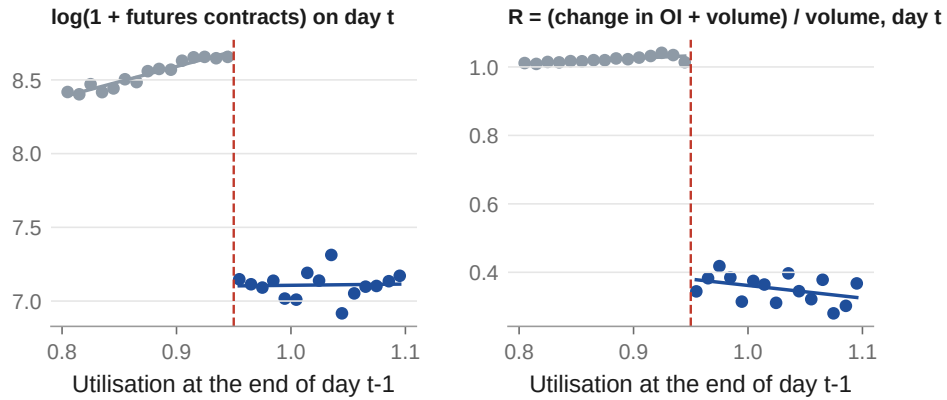


Figure 14: Futures contracts and R around 95%.

Futures -1.53 log points, R -0.66, delivery share $+5.1$ points; cash volume no jump.

The 2025 rules and “spurious” bans

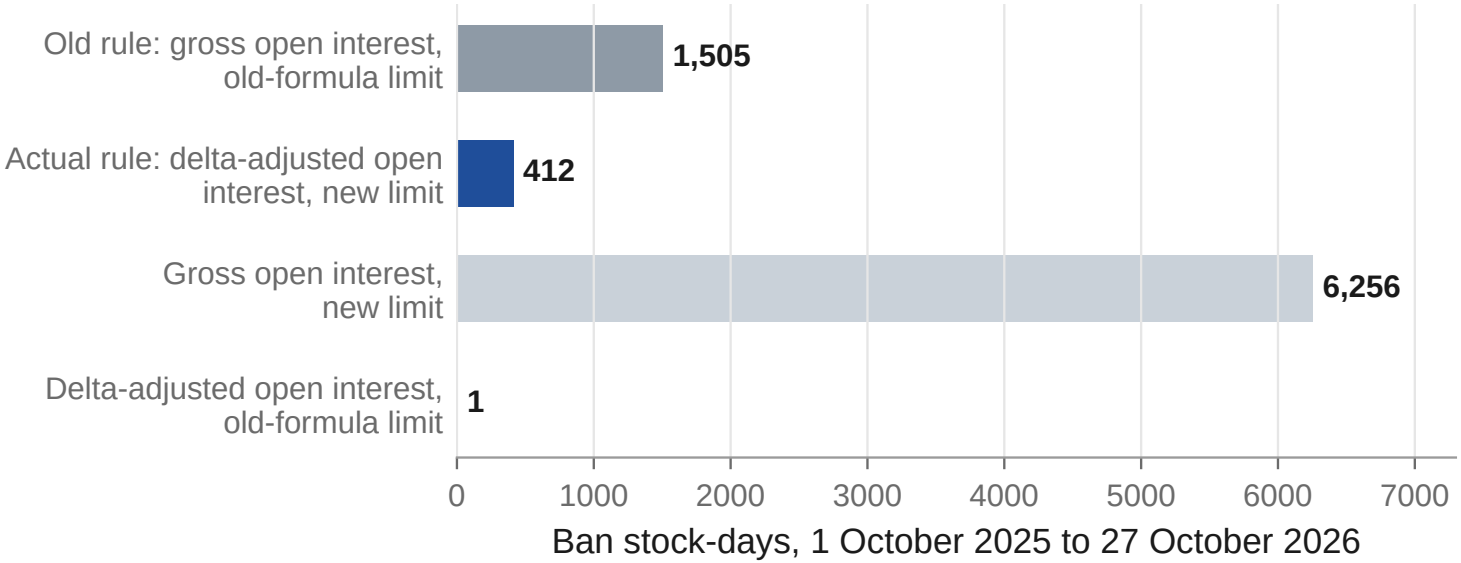


Figure 15: Ban stock-days, 1 October 2025 to 27 October 2026, under the actual and three counterfactual rules.

- Before 2025 bans came from rising open interest: only 0.4% of entries were caused by a falling limit.
- Delta-adjusted open interest removes almost all bans; the lower limit alone would have quadrupled them. Net: 412 ban days against 1,505 under the old rule.

Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The No-Arbitrage Band
7. Results II: What Sets the Band
8. Results III: Does the Ban Bind on Trading?
9. Conclusions and Stage II

Summary of the Stage I tests

Table 7: Pre-specified tests of Stage I

Test	Result	Comment
H8 trading in a ban	Supported ($p < 10^{-10}$)	Every trade closes positions under intraday checks
H3 costs and the band	Supported ($p < 10^{-10}$)	Weak against placebo dates; a common shift
H1 frictions and the band	$p = 0.002$	Across stocks only
Density at 95%	$p = 0.86$	No bunching
Band estimation	sup-F 7,471	-34.5 to $+23.9$ bp; stable over time

Primary tests (bold) Holm-corrected with the two primary tests of Stage II.

Conclusions

- The basis of Indian stock futures moves in a **wide, asymmetric band**, about -35 to $+24$ bp around each stock's median, much wider than intraday bands.
- **Costs shape the band loosely**: a new position costs 27 to 31 bp, in the outer tail; the upper edge rose after tax increases, but not unusually.
- **Frictions** widen the band across stocks.
- **The ban binds hard on trading**: futures volume -87%; every trade closes positions once checks are intraday. Enforcement matters as much as the rule.
- The 2025 rules cut ban days by about **three quarters**.

Limitations and Stage II

Limitations

- Daily data: noise inflates measured reversion
- Arbitrage positions are not observed
- Three tax events, one confounded
- Only 44 ban spells since October 2025

Stage II

- The ban and the basis: one-sided correction (H7), the level at ban start, placebo bans
- Position limits before a ban; the 2020 halving
- Arbitrage capital and the band
- Short-sale access: F&O entry (H6), anomalies, a constraint index
- Policy on position limits

Important references I

[1] Aggarwal, N. (2015). Limits to arbitrage: The case of single stock futures and spot prices. IGIDR WP-2015-010.

[2] Cattaneo, M., M. Jansson and X. Ma (2020). Simple local polynomial density estimators. *J. American Statistical Association* 115(531).

[3] De Long, J. B., A. Shleifer, L. Summers and R. Waldmann (1990). Noise trader risk in financial markets. *J. Political Economy* 98(4).

[4] Dwyer, G., P. Locke and W. Yu (1996). Index arbitrage and nonlinear dynamics between the S&P 500 futures and cash. *Rev. Financial Studies* 9(1).

[5] Gromb, D. and D. Vayanos (2010). Limits of arbitrage. *Annual Rev. Financial Economics* 2.

[6] Kolari, J. and S. Pynnönen (2010). Event study testing with cross-sectional correlation of abnormal returns. *Rev. Financial Studies* 23(11).

[7] SEBI (2025). Circular SEBI/HO/MRD/TPD-1/P/CIR/2025/79; NSE Clearing circular 119/2025.

[8] Shankar, R. L., G. Sankar and K. Kiran Kumar (2019). Mispricing in single stock futures: Empirical examination of Indian markets. *Emerging Markets Finance and Trade* 55(7).

[9] Shleifer, A. and R. Vishny (1997). The limits of arbitrage. *J. Finance* 52(1).

[10] Slivka, R., V. Shah and J. Wu (2011). Arbitrage of single stocks versus futures in India. Working paper.

[11] Ters, K. and J. Urban (2020). Estimating unknown arbitrage costs: Evidence from a 3-regime threshold vector error correction model. *J. Financial Markets* 47.

[12] Tse, Y. (2001). Index arbitrage with heterogeneous investors: A smooth transition error-correction analysis. *J. Banking and Finance* 25(10).

Acknowledgement

I thank my supervisor, Prof. Aditi Chaubal, for her guidance throughout this project, and in particular for the advice to build a formal model of what the project would test before looking at any data.

Thank You

Questions and comments are welcome

Backup: the propositions and the tests

Proposition	Statement	Tested in
1 Band	Arbitrage only outside $[-c_-, c_+]$	Stage I
2 What moves it	Costs set the edges; capital the passthrough	Stage I (H3), II (H4)
3 Existing book	The no-trade region moves with the position	Design of H3
4 One-sided	In a ban rich futures are not corrected	Stage II (H7)
5 Level effect	The ban's effect on the level has no fixed sign	Stage II
6 Trading	Each trade in a ban closes positions: $\Delta OI = -V$	Stage I (H8)

Proofs are in the appendix of the Stage I report.

Backup: the basis and time to expiry

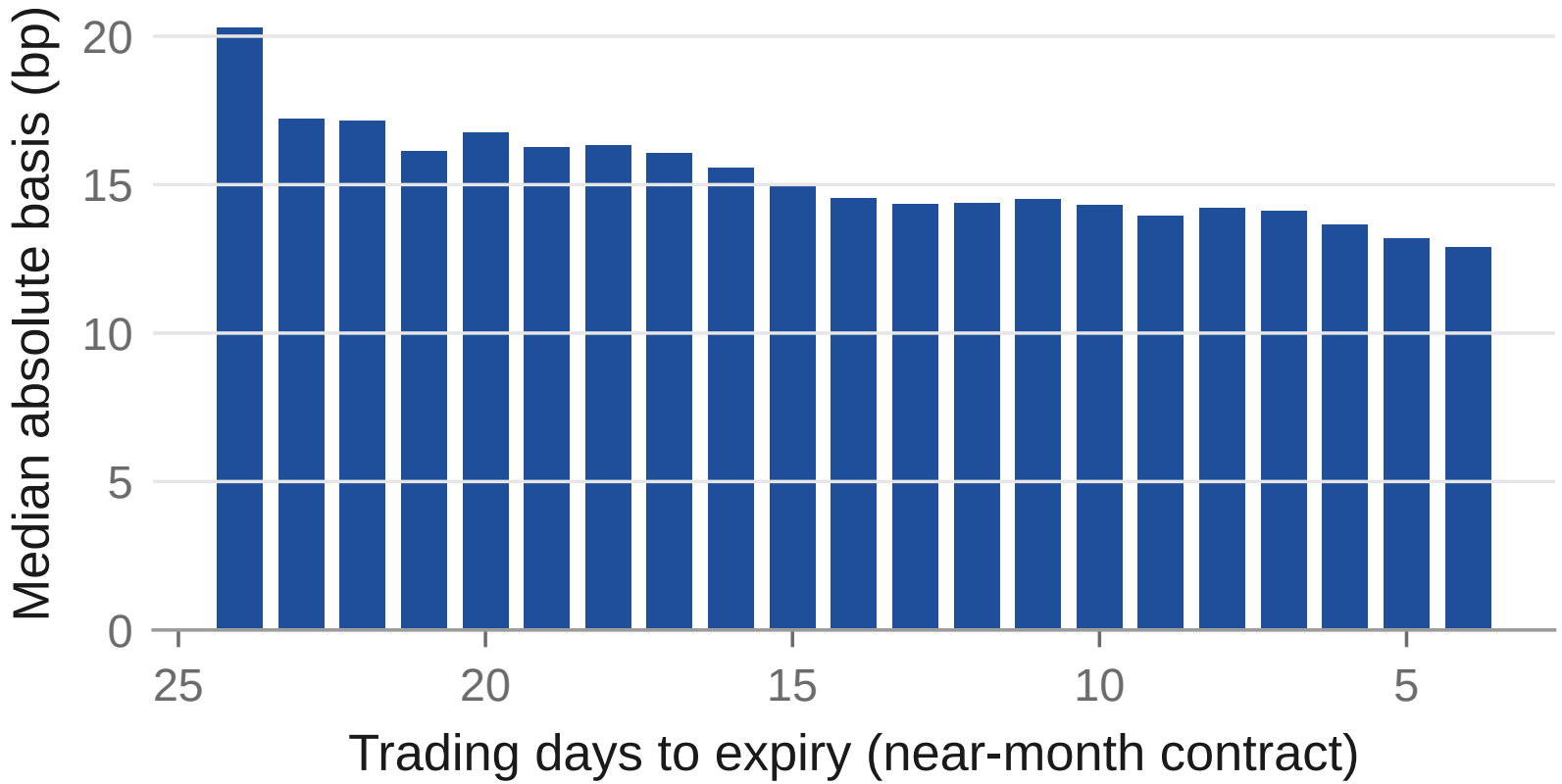


Figure 16: Median absolute basis of the near-month contract by trading days to expiry.

Backup: the October 2025 cut in the position limit

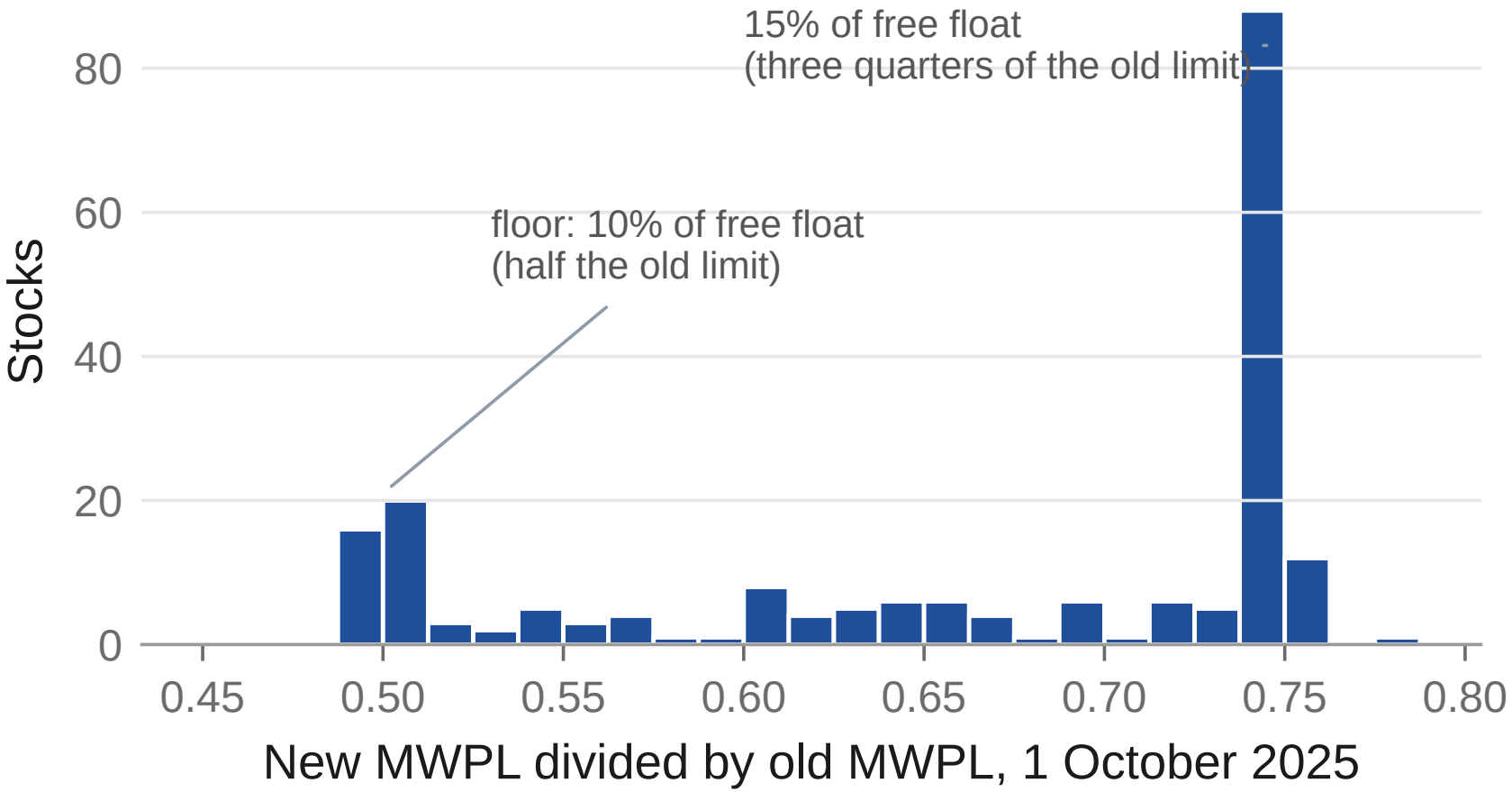


Figure 17: New MWPL divided by old MWPL, 1 October 2025 (208 stocks).

Backup: the pre-analysis plan and its log

- Frozen with a time-stamped tag on 5 November 2026, after the supervisor's go-ahead and a full set of data checks; until then no outcome was related to any treatment.
- Every detail the plan left open (sample filters, utilisation bins, placebo dates, density population) was fixed in a dated log before the test that needed it.
- Analyses added after a result (the split by enforcement period, the noise correction) are labelled as such; the pre-specified result is reported first.
- Each estimator first recovered known effects on simulated data.